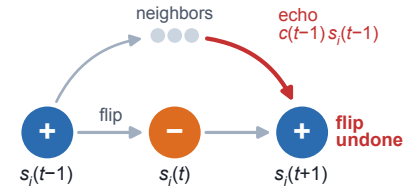
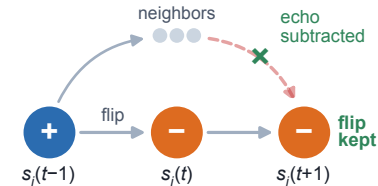


(a) Synchronous SCA: the echo

1. All spins update at once, so the neighbors respond to spin i 's old state $s_i(t-1)$ as it flips.
2. Their response returns in the spin's local field $h_i(t)$ as the echo $c(t-1)s_i(t-1)$ of strength c and pushes the spin back: 67% of all flips undo a flip one step earlier (c).

(b) Anechoic: subtract the echo

1. Each decision uses the corrected field $h_i(t) - \lambda c(t-1)s_i(t-1)$, with a gain λ .
2. c needs one population count per step, or none if set from the temperature (Onsager- κT).
3. The flip is kept: only 37% of flips are undone (c).

(c) Effect on K2000